

Firestore Integration Complete

Status: READY TO USE

Your Firestore configuration has been successfully integrated into the EnerGenius dashboard!

Configuration Details

Firestore Project Information

- **Project ID:** `rrgenergenius`
- **Project Name:** RRG EnerGenius
- **Authentication Domain:** `rrgenergenius.firebaseio.com`
- **Storage Bucket:** `rrgenergenius.firebaseio.com`
- **App ID:** `1:834259132930:web:897ce36b51872139763d1b`

Services Enabled

- ✓ **Firestore Authentication** - Anonymous sign-in ready
- ✓ **Cloud Firestore** - Real-time database configured
- ✓ **Firestore Analytics** - User analytics tracking (optional)

What Has Been Configured

1. Environment Variables (`.env`)

```
NEXT_PUBLIC_FIREBASE_API_KEY="AIzaSyAvyNxmj0yYvWP8sN-l8RQqzhS1ZfHU13U"
NEXT_PUBLIC_FIREBASE_AUTH_DOMAIN="rrgenergenius.firebaseio.com"
NEXT_PUBLIC_FIREBASE_PROJECT_ID="rrgenergenius"
NEXT_PUBLIC_FIREBASE_STORAGE_BUCKET="rrgenergenius.firebaseio.com"
NEXT_PUBLIC_FIREBASE_MESSAGING_SENDER_ID="834259132930"
NEXT_PUBLIC_FIREBASE_APP_ID="1:834259132930:web:897ce36b51872139763d1b"
NEXT_PUBLIC_FIREBASE_MEASUREMENT_ID="G-H4SNW5FGFT"
```

2. Firestore SDK Integration (`lib/firebase.ts`)

- ✓ Firestore app initialization
- ✓ Authentication module
- ✓ Firestore database module
- ✓ Analytics module (with browser-only support)
- ✓ Singleton pattern for optimal performance

3. Dashboard Application (`app/dashboard/page.tsx`)

- ✓ Real-time data synchronization
- ✓ Anonymous user authentication
- ✓ Generator metrics tracking

-  Savings calculator with cloud storage
-  Multi-tab navigation system

Next Steps to Complete Firebase Setup

REQUIRED: Enable Firebase Services

Your Firebase credentials are configured, but you must enable the services in Firebase Console:

Step 1: Enable Authentication

1. Go to [Firebase Console](https://console.firebase.google.com/project/rrgenergenius) (<https://console.firebase.google.com/project/rrgenergenius>)
2. Navigate to **Build > Authentication**
3. Click **“Get started”**
4. Select **“Anonymous”** sign-in method
5. Toggle **Enable** and click **“Save”**

Why? This allows users to access the dashboard without creating accounts while still maintaining secure, individual data storage.

Step 2: Enable Firestore Database

1. In Firebase Console, go to **Build > Firestore Database**
2. Click **“Create database”**
3. Choose **“Start in test mode”** (for now)
4. Select your preferred location (e.g., `us-central1`)
5. Click **“Enable”**

Why? This creates the real-time database where generator metrics and user settings are stored.

Step 3: Configure Firestore Security Rules

Once Firestore is enabled, update the security rules:

1. In Firestore Database, click **“Rules”** tab
2. Replace the default rules with:

```
rules_version = '2';
service cloud.firestore {
  match /databases/{database}/documents {
    // Allow users to read/write their own data
    match /artifacts/energenius-app-default/users/{userId}/{document=**} {
      allow read, write: if request.auth != null && request.auth.uid == userId;
    }

    // Deny all other access
    match /{document=**} {
      allow read, write: if false;
    }
  }
}
```

1. Click **“Publish”**

Why? This ensures each user can only access their own data, providing privacy and security.

Step 4 (Optional): Enable Analytics

Firebase Analytics is already configured but requires no additional setup. It will automatically track:

- Page views
- User engagement
- Dashboard usage patterns
- Feature adoption

Analytics data will appear in Firebase Console after users start using the dashboard.



Testing Your Firebase Integration

Test 1: Access the Dashboard

```
# Development
visit http://localhost:3000/dashboard

# Production (after deployment)
visit https://your-domain.com/dashboard
```

Expected Result:

- ☒ Loading screen appears briefly
- ☒ Dashboard loads with default data
- ☒ Navigation tabs are functional
- ☒ User ID displays in sidebar (e.g., `abc123xyz789`)

Test 2: Verify Authentication

Open browser DevTools Console (F12) and check for:

- ☒ No Firebase authentication errors
- ☒ Message: "Firebase initialized successfully"

If you see errors:

- "Firebase: Error (auth/operation-not-allowed)" → Enable Anonymous auth in Firebase Console
- "Missing or insufficient permissions" → Configure Firestore security rules

Test 3: Test Data Persistence

1. Go to **Savings Tool** tab
2. Enter values:
 - Electric Rate: `$0.20`
 - Daily Production: `15 kWh`
3. Click **"Save & Update Profile"**
4. Refresh the page
5. Go back to **Savings Tool** tab

Expected Result:

- ☒ Values are preserved after refresh
- ☒ Calculated savings display correctly

Test 4: Multi-Device Sync

1. Open dashboard in Browser 1
2. Note the **User ID** in the sidebar
3. Open dashboard in Browser 2 (same device or different)
4. Sign in with the same User ID (if auth allows)
5. Change savings data in Browser 1

Expected Result:

- ☒ Changes appear in Browser 2 within 1-2 seconds (real-time sync)

Test 5: Verify Firestore Data

1. Go to [Firebase Console](https://console.firebase.google.com/project/rrgenergenius/firestore) (https://console.firebase.google.com/project/rrgenergenius/firestore)
2. Navigate to **Firestore Database > Data**
3. Look for document structure:

artifacts → energenius-app-default → users → {userId} → appData → generatorMetrics

Expected Result:

- ☒ User documents appear after first dashboard visit
- ☒ Data updates in real-time when you change values



Data Structure

Firestore Document Path

artifacts/energenius-app-default/users/{userId}/appData/generatorMetrics

Document Schema

```
{
  "generator": {
    "health": 98,
    "currentCharge": 75,
    "dailyUsageKWH": 8.5,
    "temperature": 35,
    "runtimeHours": 1250
  },
  "savings": {
    "currentRate": 0.15,
    "dailySolarProduction": 10,
    "potentialCreditPerMonth": 45.60
  }
}
```



Security & Privacy

Authentication Security

- **Anonymous Auth:** Each user gets a unique, persistent ID
- **No Personal Data:** No email, password, or identifying information required

- **Session Persistence:** Users maintain access across devices with same browser

Data Privacy

- **User Isolation:** Each user can only access their own data
- **Firestore Rules:** Server-side enforcement prevents unauthorized access
- **No Shared Data:** Generator metrics are private to each user

API Key Security

- **Public API Keys:** Firebase API keys are safe to expose (domain-restricted)
- **Rate Limiting:** Firebase automatically prevents abuse
- **Quota Management:** Monitor usage in Firebase Console



Troubleshooting

Issue: “Firebase is not configured” Warning

Cause: Environment variables not loaded or missing

Solution:

1. Verify `.env` file exists in `/nextjs_space/`
2. Check all variables start with `NEXT_PUBLIC_`
3. Restart dev server: `yarn dev`
4. Hard refresh browser: `Ctrl+Shift+R` (or `Cmd+Shift+R` on Mac)

Issue: Authentication Error

Cause: Anonymous sign-in not enabled

Solution:

1. Go to Firebase Console → Authentication
2. Enable “Anonymous” provider
3. Save and retry

Issue: “Permission Denied” Errors

Cause: Firestore security rules not configured

Solution:

1. Go to Firebase Console → Firestore Database → Rules
2. Copy the rules from **Step 3** above
3. Publish rules
4. Wait 30 seconds for propagation

Issue: Data Not Syncing

Cause: Firestore listener not attached or network error

Solution:

1. Check browser console for errors
2. Verify Firestore is enabled in Firebase Console
3. Check network tab for `firestore.googleapis.com` requests
4. Ensure User ID is displayed in sidebar

Issue: Analytics Not Tracking

Cause: Analytics requires opt-in or browser blocks

Solution:

- Analytics works automatically in most browsers
 - Ad blockers may prevent tracking (expected behavior)
 - Check Firebase Console → Analytics dashboard (data appears after 24 hours)
-



Monitoring & Maintenance

Firebase Console URLs

- **Project Overview:** <https://console.firebase.google.com/project/rrgenergenius>
- **Authentication:** <https://console.firebase.google.com/project/rrgenergenius/authentication>
- **Firestore Database:** <https://console.firebase.google.com/project/rrgenergenius/firestore>
- **Analytics:** <https://console.firebase.google.com/project/rrgenergenius/analytics>
- **Usage & Billing:** <https://console.firebase.google.com/project/rrgenergenius/usage>

Key Metrics to Monitor

1. **Authentication**

- Daily active users
- Sign-in methods used
- Authentication errors

2. **Firestore**

- Read/Write operations
- Document count
- Storage size

3. **Analytics**

- Page views on /dashboard
- Time spent per tab
- Feature engagement

Free Tier Limits (Spark Plan)

- **Authentication:** 10,000 verifications/month
- **Firestore:** 50,000 reads/day, 20,000 writes/day
- **Storage:** 1 GB
- **Analytics:** Unlimited events

When to Upgrade:

- If you exceed free tier limits
 - If you need phone authentication
 - If you require extended Firestore storage
-

Future Enhancements

Phase 1: IoT Integration

- Connect to actual solar generator hardware
- Real-time telemetry via MQTT/WebSocket
- Live battery voltage and temperature readings

Phase 2: Advanced Authentication

- Email/Password authentication
- Google Sign-In integration
- Multi-user household accounts

Phase 3: Document RAG System

- Upload generator manuals to Cloud Storage
- AI-powered document search
- Smart warranty lookups

Phase 4: Alerts & Notifications

- Firebase Cloud Messaging (push notifications)
- Email alerts for low battery or maintenance
- SMS notifications for critical events

Verification Checklist

Before considering the Firebase integration complete, verify:

- ☐ Firebase Console accessible at console.firebase.google.com
- ☐ Anonymous authentication enabled
- ☐ Firestore database created
- ☐ Security rules configured and published
- ☐ Environment variables set in `.env`
- ☐ Dashboard loads without errors
- ☐ User ID displays in sidebar
- ☐ Data persists after page refresh
- ☐ Real-time sync works between tabs
- ☐ No console errors in browser DevTools

Support

Firebase Documentation

- [Firebase Console](https://console.firebase.google.com) (<https://console.firebase.google.com>)
- [Firebase Docs](https://firebase.google.com/docs) (<https://firebase.google.com/docs>)
- [Firestore Getting Started](https://firebase.google.com/docs/firestore/quickstart) (<https://firebase.google.com/docs/firestore/quickstart>)
- [Firebase Authentication](https://firebase.google.com/docs/auth) (<https://firebase.google.com/docs/auth>)

EnerGenius Support

- Email: support@renewableresourcegroup.net
 - Dashboard Issues: Check `/dashboard` for error messages
 - Technical Documentation: See `README_DASHBOARD.md`
-

Congratulations!

Your Firebase-powered customer dashboard is ready to revolutionize how EnerGenius customers monitor and manage their solar generators!

EnerGenius - Power That Thinks Ahead

by Renewable Resource Group LLC